

Ravindranath Tagore

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PROFESSIONAL SUMMARY

Software Engineer experienced in building AI voice-agent platforms, backend services, RAG-based knowledge systems, workflow orchestration, cloud infrastructure, and observability systems. Skilled in Python, JavaScript, FastAPI, React, PostgreSQL, Redis, AWS, Docker, Temporal, Celery, Prometheus, Grafana, and distributed systems, with strong experience improving latency, reliability, scalability, and production debugging.

EDUCATION

Dayananda Sagar Academy of Technology and Management

Bachelor of Engineering in Information Science and Engineering; CGPA: 8.44

Bengaluru, Karnataka

Graduated 2026

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL

Backend: FastAPI, REST APIs, WebSockets, Distributed Systems, System Design

AI/LLM Systems: Knowledge Base Architecture, Prompt Optimization, LLM Observability, STT/TTS Integration, AI Voice Agents

Workflows: Temporal, Celery, Celery Beat, Activity Workers, Retry Handling, Circuit Breakers

Cloud/DevOps: AWS EC2, AWS RDS, Auto Scaling, ALB, Docker, Linux, Nginx, systemd, CI/CD

Databases/Cache: PostgreSQL, Redis, Supabase, Session Management, Query Debugging

Monitoring: Prometheus, Grafana, Loki, Custom Metrics, Latency Tracking, Production Log Analysis

Security: JWT Revocation, Audit Logging, Idle Timeout, Concurrent Session Control, Client IP Tracking

EXPERIENCE

Software Engineer Intern

2026 - Present

Agentic Universe (Formi)

Bengaluru, Karnataka

- Built and shipped an Enterprise Knowledge Base and RAG system across multiple enterprise AI voice agents, including Cashify and Generali Insurance, enabling knowledge-base upload, compile, publish, versioning, agent-level enablement, and real-time runtime retrieval.
- Reduced large AI voice-agent prompts by 45-50% by moving FAQs, objections, policy rules, compliance guardrails, and branch/location logic into a structured RAG knowledge base.
- Achieved 63.5 ms average RAG retrieval latency with p90 of 95 ms and p95 of 97 ms, while validating retrieval quality through 162 UAT test cases for multilingual English-Hindi voice flows.
- Migrated dialler orchestration toward Temporal workflows, replacing fragile scheduling flows with workflow-based dispatch, activity workers, quota governance, retries, circuit-breaker handling, and operational monitoring.
- Supported staged Temporal rollout for production traffic, including 20% to 50%+ traffic shifts, workflow monitoring, failure RCA, Redis circuit debugging, and live production log analysis.
- Architected and deployed AWS-based backend infrastructure supporting 100-900+ concurrent AI voice calls, ensuring 0 active call drops through lifecycle-aware graceful instance draining and connection-safe scaling.
- Reduced end-to-end AI voice-agent latency by 30-40% by resolving streaming and backend bottlenecks, eliminating stuttering and improving real-time response consistency.
- Built end-to-end observability using Prometheus, Grafana, Loki, systemd logs, Redis, PostgreSQL, and custom metrics for active calls, RAG latency, LLM failures, TTS latency, Celery workers, and Temporal workflows.
- Implemented InfoSec session controls including idle timeout, concurrent login limits, Redis-based JWT revocation, audit logging, client IP tracking, and session termination policies.
- Improved post-call processing reliability by narrowing database session scope around long-running LLM calls, preventing idle transaction timeout failures during 60-200 second LLM operations.
- Debugged staging and production issues across PostgreSQL, Redis, Celery, Temporal, Nginx, AWS RDS, EC2, Docker, and Linux systemd using repeatable live-log, health-check, and RCA workflows.

PROJECTS

Slates.me - Canvas Collaboration Platform | React, Supabase, Real-Time Collaboration

[Live Site](#)

- Built a real-time collaborative canvas platform using React and Supabase for persistent storage and synchronized user interactions.
- Implemented shared canvas functionality allowing users to add, move, and manage notes and files including images, videos, and PDFs.
- Developed responsive UI with gesture support, auto-save functionality, and cross-device accessibility.

HOPS - Hybrid Orbital Prediction System | Python, Flask, Machine Learning, Space Data

GitHub

- Developed a hybrid satellite orbital lifetime prediction system combining the JB2008 physics model with a Random Forest algorithm using 100 decision trees.
- Achieved 98.5% error reduction by training on 32,413 TLE records from 6 verified reentry satellites including UARS, ROSAT, Tiangong-1, Tiangong-2, Compton GRO, and Phobos-Grunt.
- Validated using Leave-One-Satellite-Out cross-validation, achieving 11.7% average improvement on unseen satellites and up to 50% error reduction for similar orbital profiles.
- Built a Flask web application with real-time Celestrak and Space-Track integration and transparent SEEN/UNSEEN accuracy indicators.

CERTIFICATIONS

Microsoft Certified: Azure Fundamentals (AZ-900)

June 2025

- Gained foundational knowledge of cloud concepts, Azure services, security, governance, pricing, and cloud deployment models.

ACHIEVEMENTS

- Received Local Award at NASA Space Apps Challenge 2024, DSU Local Host, for innovation and impact among 509+ teams.
- Won 1st Prize in Inter-College Innovative Idea and Model Competition at Dayananda Sagar Academy of Technology and Management, Bengaluru.
- Secured Second Runner-Up in State-Level Project Exhibition 2025 at Bangalore Institute of Technology.
- Secured Second Runner-Up in Inter-College Frontend Development Competition at Dayananda Sagar Academy of Technology and Management, Bengaluru.
- Won Consolation Prize in Altair Data Science Contest, organized by Department of Management Sciences, IIT Kanpur using Altair RapidMiner.

EXTRACURRICULAR ACTIVITIES

- Core Member of Innovisions Technical Club, ISE Department, contributing to student-led technical initiatives, project activities, and innovation-focused events.